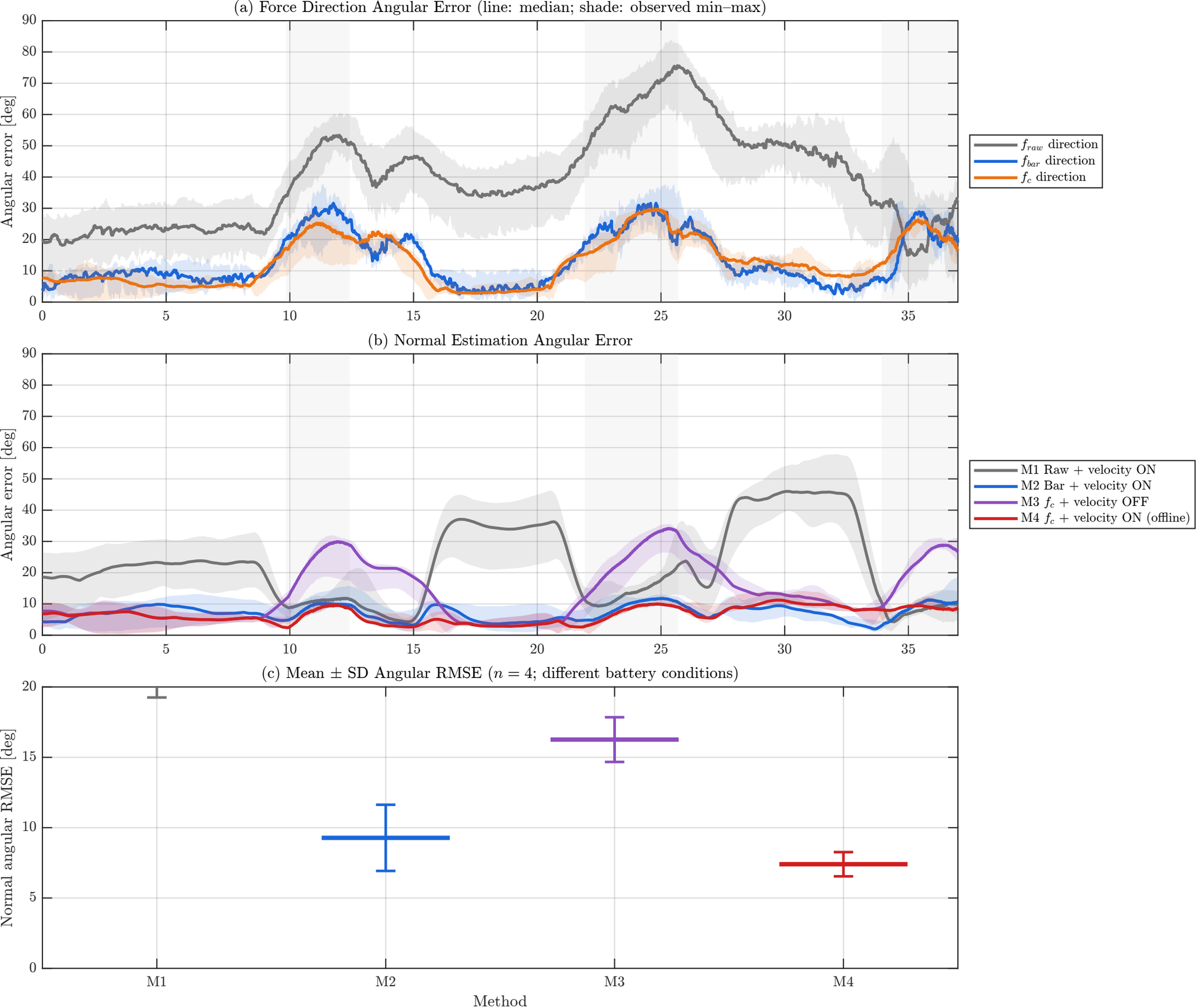

(a) Force Direction Angular Error (line: median; shade: observed min–max)
Angular error [deg]
f_{raw} direction
f_{bar} direction
f_c direction
(b) Normal Estimation Angular Error
Angular error [deg]
M1 Raw + velocity ON
M2 Bar + velocity ON
M3 f_c + velocity OFF
M4 f_c + velocity ON (offline)
(c) Mean ± SD Angular RMSE ($n = 4$; different battery conditions)
Normal angular RMSE [deg]
M1
M2
M3
M4
Method